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Scientists can now study the **event horizons** of black holes
科学家如今可以研究黑洞的事件视界

【 Para.1 】

A long time ago, in a galaxy far, far away—somewhere in the general direction of the **constellation** of Leo—two black holes crashed together and **merged**. About 1.3bn years later, on January 14th 2025, a **blip** appeared in the detectors at the Laser **Interferometer Gravitational-Wave Observatory** (LIGO), an **instrument** based in Louisiana and Washington state.

很久很久以前，在一个遥远的星系中，朝着狮子座的方向，有两个黑洞碰撞并最终合并在一起。13亿年后，即2025年1月14日，美国激光干涉引力波天文台（称LIGO）的探测器中出现了一个微弱的信号。LIGO是一套位于美国路易斯安那州和华盛顿州的引力波探测设备。

July 4TH-10TH | Science & technology

文章导读：

科学家借助 LIGO 捕捉到 13 亿年前双黑洞合并产生的引力波。研究团队利用该清晰信号，首次观测到黑洞事件视界，观测结果完全符合广义相对论预测。新技术探明了视界附近的光线红移与时空扭曲现象。由于黑洞奇点会让广义相对论失效，此次突破也为验证量子引力效应、调和两大物理理论矛盾提供了全新途径。

event horizon n. /ɪ'vent hə'raɪzn/ the outer edge of a black hole, beyond which nothing, including light, can escape 视界，视界（即黑洞表面）

constellation n. /kən'stə'leɪʃn/ any of the groups of stars in the sky that seem from earth to form a pattern and have been given names 星座

merge v. /mɜːdʒ/ (~ with/into sth) (~ A with B) (~ A and B together) to combine or make two or more things combine to form a single thing (使) 合并，结合，并入【考研英语大纲词汇】
The company was formed by **merging** three smaller firms. 公司是由三家小公司合并组成的。

blip n. /blɪp/ a bright light flashing on the screen of a piece of equipment, sometimes with a short high sound 目标（显示）标志（仪表屏幕上的闪亮光点，有时伴有短促尖声）

interferometer n. /ɪntə'fɜːrəmə'tɜː(r)/ any acoustic, optical, or microwave instrument that uses interference patterns or fringes to make accurate measurements of wavelength, wave velocity, distance, etc. 干涉仪

gravitational adj. /grævɪ'teɪʃnəl/ connected with or caused by the force of gravity 引力的；重力引起的【考研英语大纲词汇】

If a spacecraft travels faster than 11 km a second, it escapes the Earth's **gravitational** pull. 如果一架宇宙飞船飞行超过每秒11公里，它就能摆脱地球引力

observatory n. /əb'zɜːvətɹi/ a special building from which scientists watch the stars, the weather, etc. 天文台；天文观测站；气象台

instrument n. /'ɪnstɾəmənt/ a device used for measuring speed, distance, temperature, etc. in a vehicle or on a piece of machinery (车辆、机器的) 仪器，仪表【考研英语大纲词汇】

pulse n. /pʌls/ a single short increase in the amount of light, sound or electricity produced by a machine, etc. 脉冲【考研英语大纲词汇】

ripple n. /'rɪpl/ a small wave on the surface of a liquid, especially water in a lake, etc. 波纹；细浪；涟漪【考研英语大纲词汇】

fabric n. /'fæbrɪk/ (formal) the structure or parts of something 结构【考研英语大纲词汇】

emit v. /i'mɪt/ (formal) to send out sth such as light, heat, sound, gas, etc. 发出，射出，散发（光、热、声音、气等）【考研英语大纲词汇】

The metal container began to **emit** a clicking sound. 金属容器开始发出咔嗒咔嗒的声音。

【 Para.2 】

The blip marked the arrival of the **pulse** of gravitational waves—**ripples** in the **fabric** of the universe—**emitted** as the black holes **collided**. In a paper published in Nature on June 24th, a team led by Sizheng Ma of Perimeter Institute for Theoretical Physics, in Ontario, report that they have used those waves to **glimpse** one of nature's strangest phenomena: the “event horizons” that surround black holes. 这个微弱信号标志着一束引力波脉冲的到来。引力波是宇宙结构中的涟漪，由两个黑洞碰撞时释放。6月24日，发表在《自然》杂志的一篇论文中，加拿大安大略省理论物理周界研究所的马思正（Sizheng Ma）领导下的研究团队报告称，他们利用这些引力波窥见了自然界最奇特的现象之一，环绕黑洞的“事件视界”。

【 Para.3 】

An event horizon marks the point beyond which gravity so **warps** spacetime that nothing, not even light, can escape. An object that crosses the horizon is cut off from the rest of the universe, with all possible roads leading to the black hole's centre. Event horizons are **invisible**, although their effects can often be seen when a black hole is swallowing hot, glowing gas. Physicists nevertheless know a great deal about how they work, thanks to Albert Einstein's theory of general relativity. But physics is an **empirical** science as well as a theoretical one. Dr Ma thought that gravitational waves might offer a way to check that Einstein's theory does indeed match reality at the boundary of a black hole.

collide v. /kə'laɪd/ if two people, vehicles, etc. collide, they crash into each other; if a person, vehicle, etc. collides with another, or with sth that is not moving, they crash into it 碰撞；相撞【考研英语词汇】

The car and the van **collided** head-on in thick fog. 那辆小轿车和货车在浓雾中迎面相撞。

glimpse v. /glɪmps/ to see sb/sth for a moment, but not very clearly 瞥见；看一眼【考研英语词汇】

He'd **glimpsed** her through the window as he passed. 他路过时透过窗户瞥见了她。

SYN catch sight of, spot

warp v. /wɔ:p/ to become, or make sth become, twisted or bent out of its natural shape, for example because it has become too hot, too damp, etc. (使) 扭曲，弯曲，变形

The window frames had begun to **warp**. 窗框已经开始变形。

invisible adj. /ɪm'vɪzəbl/ (~ to sb/sth) that cannot be seen 看不见的；隐形的【考研英语词汇】

empirical adj. /ɪm'pɪrɪkl/ based on experiments or experience rather than ideas or theories 以实验（或经验）为依据的；经验主义的【考研英语词汇】

property n. /'prɒpəti/ (formal) a quality or characteristic that sth has 性质；特性【考研英语词汇】

interpret v. /ɪn'tɜ:pɪt/ (~ sth as sth) to decide that sth has a particular meaning and to understand it in this way 把...理解为；领会【考研英语词汇】

The data can be **interpreted** in many different ways. 这份资料可以从多方面解读。

binary adj. /'bɪnəri/ based on only two numbers; consisting of two parts 仅基于两个数字的；二元的；由两部分组成的【考研英语词汇】

A **binary** star system consists of two stars orbiting a common center of mass. 双星系统由两颗绕共同质心旋转的恒星组成。

decipher v. /dɪ'saɪfə(r)/ to succeed in finding the meaning of sth that is difficult to read or understand 破译，辨认（难认、难解的东西）

I'm still no closer to **deciphering** the code. 我破译这个密码还是没有进展。

unprecedented adj. /ʌn'presɪdəntɪd/ that has never happened, been done or been known before 前所未有的；空前的；没有先例的【考研英语词汇】

bizarre adj. /bɪ'zɑ:(r)/ very strange or unusual 极其怪诞的；异乎寻常的【考研英语词汇】

immense adj. /ɪ'mens/ extremely large or great 极大的；巨大的【考研英语词汇】

rotate v. /rəʊ'teɪt/ (~ about/around sth) to move or turn around a central fixed point; to make sth do this (使) 旋转，转动【考研英语词汇】

Rotate the wheel through 180 degrees. 将方向盘转动180度。

tug v. /tʌg/ to pull sth hard in a particular direction (朝某一方向用力) 拉，拖，拽【考研英语词汇】

He **tugged** the hat down over his head. 他把帽子往下拉了拉遮住脸。

passage n. /'pæsɪdʒ/ (the ~ of time literary) the process of time passing (时间的) 流逝，推移【考研英语词汇】

事件视界标识出引力界限，一旦跨过该边界，引力对时空的扭曲将剧烈到任何东西、甚至连光都无法逃脱。任何穿越事件视界的物体都会与宇宙其他部分隔绝，所有可能的路径最终都通向黑洞中心。事件视界本身不可见，但当黑洞吞噬炽热、发光的气体时，人们往往能够观察到它所产生的影响。得益于阿尔伯特·爱因斯坦（Albert Einstein）的广义相对论，物理学家已经对事件视界的运行方式有了大量了解。然而，物理学既是一门理论科学，也是一门经验科学。马博士认为，引力波或许能够提供一种方法，用来检验爱因斯坦的理论是否确实符合黑洞边界的现实情况。

【 Para.4 】

General relativity predicts that as two black holes collide and merge, the **properties** of the new hole's event horizon should be **imprinted** on the gravitational waves released. To test this, Dr Ma teamed up with physicists from LIGO, who are used to **interpreting** the complex signals that show up in their detectors. Since the wave detected last year is the clearest from a **binary** merger so far detected, the team focused their attention there. Sure enough, they were able to **decipher** the part of the signal in question. And it matched their theoretical prediction exactly.

广义相对论预测，当两个黑洞发生碰撞并合并时，新形成黑洞的事件视界特性应当会被印刻在释放出的引力波中。为了验证这一点，马博士与 LIGO 的物理学家展开合作。这些科学家长期致力于解读探测器中出现的复杂信号。由于去年探测到的这次引力波是迄今为止观测到的双黑洞合并事件中最清晰的一次，研究团

rumble v. /'rʌmbəl/ to make a long deep sound or series of sounds 发出持续而低沉的声音；发出隆隆声

probe v. /prəʊb/ if you probe into something, you ask questions or try to discover facts about it 调查；探寻【考研大纲词汇】

The more they **probed** into his background, the more inflamed their suspicions would become. 他们越深入调查他的背景，对他的怀疑就会越强烈。

nonsensical adj. /nɒn'sensɪkl/ ridiculous; with no meaning 荒谬的；无意义的

hint n. /hɪnt/ something that suggests what will happen in the future 征兆；迹象【考研大纲词汇】

quantum mechanics n. /kwɒntəm mə'kæniks/ the branch of physics that deals with the behaviour of matter and energy at the atomic and subatomic level, based on the principle that energy is emitted in discrete packets (quanta) 量子力学

feat n. /fi:t/ (approving) an action or a piece of work that needs skill, strength or courage 技艺；武艺；功绩；英勇事迹【考研大纲词汇】

The tunnel is a brilliant **feat** of engineering. 这条隧道是工程方面的光辉业绩。

elude v. /i'lu:d/ if sth eludes you, you are not able to achieve it, or not able to remember or understand it 使达不到；使不记得；使不理解【考研大纲词汇】

Finally he remembered the tiny detail that had **eluded** him the night before. 他终于想起了前一天晚上想不起来的细节。

incompatibility n. /ɪnkəm'pætə'bɪləti/ the state of not being able to exist or work with another person or thing because of basic differences 不相容，不和谐【考研大纲词汇】

linger v. /'lɪŋgə(r)/ (~ on) to continue to exist for longer than expected 继续存留；缓慢消失【考研大纲词汇】

The faint smell of her perfume **lingered** in the room. 房间里仍飘溢着她那淡淡的香水味。

➤ 今日句子分析

Para. 6 Most physicists see that as a hint that some new, more fundamental theory remains to be discovered—probably one that combines gravity with quantum mechanics, a feat that has eluded physicists for decades. 大多物理学家认为，这表明目前仍有某种全新的、更基础的理论尚待发现——很可能是一种能够将引力与量子力学结合起来的理论。而实现这一目标，正是物理学家数十年来始终未能解决的难题。



队将分析重点放在了这一信号上。果然，他们成功解析出了其中关键部分的信号，而这一结果与他们基于理论做出的预测完全一致。

【 Para.5 】

The team's new technique provides an **unprecedented** view of the **bizarre** environment just outside an event horizon. Here light loses energy as it attempts to climb out of the black hole's **immense** gravitational field. That makes objects approaching the horizon appear redder and redder to outside observers. As the black hole **rotates** it **tugs** the fabric of spacetime around with it, warping distances and the **passage** of time. The physics of both of these effects are captured in the **rumble** of the gravitational waves, allowing physicists to **probe** gravity at its most extreme.

研究团队提出的新技术，向人类展示了事件视界外部奇异环境中前所未见的情况。光在试图逃离黑洞巨大的引力场时会损失能量。对于外部观察者而言，接近事件视界的物体会显得越来越红。与此同时，黑洞在自转时会拖拽周围的时空结构，使距离和时间流逝方式发生扭曲。这两种效应所涉及的物理规律都会被记录在引力波的隆隆声中，从而使物理学家得以探测处于最极端状态下的引力。

【 Para.6 】

Or nearly its most extreme. Hidden behind the event horizon, at the heart of the black hole, is an even stranger region: the singularity. There the gravitational field is so intense that Einstein's theory starts to give **nonsensical**

定语从句（修饰 one）			
that	combines	gravity	with quantum mechanics,
主语	谓语	宾语	状语
a feat			
同位语			
定语从句（修饰 feat）			
that	has eluded	physicists	for decades.
主语	谓语	宾语	时间状语

主句：

主语：Most physicists；系语：see；（常见同义用词：regard/view/ consider/ perceive都可表“认为、理解、把……视为”。）表语：that；（代词，指代前文广义相对论在奇点处失效这一事实。非从句引导词。）表语补足语：as a hint；

同位语从句：（修饰hint，但不是定语从句，而是同位语从句，解释hint的具体内容是什么。区分二者最简单的方法是看引导词that在从句中有没有成分，如果that在从句中充当主、宾、表等，就是定语从句，例如The hint that he gave was useful. that作宾语。而在同位语从句中，从句本身结构完整，that只是连接词。对比本文a hint that some new theory remains to be discovered... that没有代替任何对象，some new theory remains to be discovered本身结构完整，that在其中不充当任何语法成分。）

引导词：that；主语：some new, more fundamental theory；（some表示“某一”，故theory是单数。）

谓语：remains；状语：to be discovered；（除了be动词+adj.表状态，remain也是常用词，且有种延续到现在的意味，如remain unsolved悬而未决，remain unknown仍然未知。）

同位语（one等同于前文所指的theory，翻译时还是要指明“一种……的理论”。）：probably one；（本文破折号引导的是短语，但也可用来引导句子，一般表示补充说明、插入解释或是突然转折。）

定语从句（修饰one）：

主语：that；谓语：combines；宾语：gravity；状语：with quantum mechanics；同位语：a feat；

定语从句（修饰feat）：

主语：that；谓语：has eluded；宾语：physicists；时间状语：for decades；（重点积累elude用法，是名词做主语，因此要表达某人未能解决/被某事难倒，可以像原文一样用名词而非人做主语，更符合英文的表达特点。）

results. Most physicists see that as a **hint** that some new, more fundamental theory remains to be discovered—probably one that combines gravity with **quantum mechanics**, a **feat** that has **eluded** physicists for decades.

或者说，让他们得以探测到接近最极端状态下的引力。在事件视界之后、黑洞的核心位置，还隐藏着一个更加陌生的区域：奇点。在那里，引力场强大到连爱因斯坦的理论都无法推导出有意义的结果。大多数物理学家认为，这表明目前仍有某种全新的、更基础的理论尚待发现——很可能是一种能够将引力与量子力学结合起来的理论。而实现这一目标，正是物理学家数十年来始终未能解决的难题。

【 Para.7 】

Event horizons are as close as the universe allows physicists to get to singularities. That makes them “very interesting from the point of view of understanding the **incompatibility** between quantum mechanics and general relativity”, says Nicolas Yunes, a physicist at the University of Illinois Urbana-Champaign, who was not involved in the new study. He hopes that “whispers of quantum-gravitational effects” may **linger** near the event horizon. Now it may be possible to check.

事件视界是宇宙允许物理学家接近奇点的极限距离。伊利诺伊大学厄巴纳-香槟分校的物理学家尼古拉斯·尤尼斯（Nicolas Yunes）表示，从理解“量子力学与广义相对论之间的不相容

➤ 写作积累

1. (Para. 3) An object that crosses the horizon is **cut off** from the rest of the universe, with all possible roads leading to the black hole's centre.

本句中的 cut off 表示 “severing something, interrupting a process, or establishing a limit or deadline”，即“隔绝，阻断”，例如，One cannot live cut off from society. 一个人脱离社会就不能生活下去。此外，还有以下表达可以表示类似的含义：

- ♦ **shut off idiom** the cessation of a flow (like water, gas, or electricity) or the device used to stop that flow 切断，停止(...的供应)

The State Water Project has shut off all supplies to farmers. 州水利工程项目已切断了对农场主的所有供水。

- ♦ **beyond the reach of idiom** something is too far away to grab 超出了...的范围，遥不可及，鞭长莫及

The scientist's theory was beyond the reach of God faith. 这位科学家的理论超越了上帝信仰的范畴。

2. (Para. 4) To test this, Dr Ma teamed up with physicists from LIGO, who **are used to** interpreting the complex signals that show up in their detectors.

本句中的 be used to (doing) 表示 “to be familiar or comfortable with something because you have done it many times”，即“习惯于，对某事已经熟悉，不觉得陌生”，例如，I'm used to getting up early. 我习惯早起。We're used to the noise from the traffic now. 现在我们已经适应车辆往来的噪音了。此外，还有以下习语可以表示类似的含义：

- ♦ **be well versed in idiom** having a lot of knowledge about something 精通、熟谙、通晓

The researchers are well versed in analyzing gravitational-wave data. 研究人员精通分析引力波数据。

- ♦ **know one's way around idiom** being very familiar with a place, tool, or subject 熟悉某事；知道如何处理

As a seasoned traveler, Sarah really knows her way around airports and can guide us effortlessly. 作为一名经验丰富的游客，萨拉对机场非常熟悉，可以毫不费力地带路。

性的角度来看，事件视界非常有趣”。尤尼斯并未参与这项新研究。他希望，在事件视界附近可能仍然残留着“量子引力效应的微弱痕迹”。如今，检验这类残迹或许成为了可能。